QUESTION THREE (13 MARKS)

a) State ONE disadvantage of using an array to implement Queues. [1 mark]
b) Differentiate between merge and sort operations [2 marks]
c) Use quick sort method to sort the following array. [4 marks]

24 56 47 35 10 90 82 31

d) The efficiency of an algorithm can be explained in terms of time and space. Support this statement. [6 marks]

QUESTION FOUR (13 MARKS)

a) Describe any of the properties of binary search algorithms. [1 mark]
b) Describe the following notations as used in time complexity of an algorithm. [3 marks]
   i. Big O
   ii. Theta
   iii. Omega
c) Study the sequence of numbers below and attempt the questions that follow.

45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.

   i. Make a Binary Search Tree [3 marks]
   ii. Traverse the tree in preorder [3 marks]
   iii. Traverse the tree in postorder [3 marks]

QUESTION FIVE (13 MARKS)

a) Describe the following terms as used in a tree data structure [3 marks]
   i. Root
   i. Leaf
   ii. Parent
b) Describe the following terms as used in data structures and algorithms. [4 marks]
   i. Recursion
   ii. Heap
c) Convert the following infix expressions into prefix and postfix expressions [6 marks]
   i. A-(B-(C-D))
   ii. A-B-C-D
   iii. (A+B)*(C/(D-E))

QUESTION SIX (13 MARKS)

a) Explain the following types of graphs. [3 marks]

i. Mixed
ii. Undirected
iii. Directed

b) Differentiate between double ended queue and circular linked list. Use illustrations. [4 Marks]

c) A queue is implemented using ADT list to presents its items. Discuss the efficiency of the queue insertion and deletion operations when the ADT list implementation is:

i. Array based [3 marks]
ii. Pointer based [3 marks]

QUESTION SEVEN (13 MARKS)

a) Explain three strategies that divide and conquer algorithms use to solve problems [3 marks]

b) Describe four real world application of algorithms [4 marks]

c) Write Pseudo code for each of the following operations of linked lists. [6 marks]

i. Inserting a node
ii. Traversing a linked list

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UNIVERSITY EXAMINATIONS
2023/2024 ACADEMIC YEAR

SECOND YEAR **FIRST** SEMESTER EXAMINATION

FOR THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY/COMPUTER SCIENCE/SCIENCE WITH EDUCATION & BSC

COURSE CODE: BIT 216

COURSE TITLE: INTRODUCTION TO NETWORKING

DATE: 11TH DECEMBER, 2023 TIME: 8.00 – 10.00 AM

INSTRUCTION TO CANDIDATES

SEE INSIDE

- Terrestial
- Satelite
- Infrared

10 bares
10 – 10 mbps
bare - max, segment length
5 or 2 - ?
bare - baseband signaling

4751b

INSTRUCTIONS:

Answer **ALL** questions in section A and **ANY TWO** from section B

SECTION A (30 MARKS) –Compulsory

Question One (30 marks)

a. A company would like to connect their computers to the Local Area Network (LAN).

i. Describe the term Local Area Network **(2 marks)**

ii. Give any TWO suitable topologies for the LAN and, for each one, draw a labelled diagram to show its structure. **(6 marks)**

iii. State any THREE items of hardware and/or software required if the LAN has to connect to the Internet. **(3 marks)**

b. There are ten employees in a local estate agent's office. Each employee uses a networked PC on a Local Area Network (LAN).

i. Give **FOUR** benefits to the office of using a network rather than stand-alone PCs. **(4 marks)**

ii. The office is part of a national chain connected together over a Wide Area Network (WAN). Explain the difference between a LAN and a WAN. **(4 marks)**

c. Describe **FOUR** goals of computer security **(8 marks)**

d. What does VOIP stand for and what is this network technology? **(3 marks)**

SECTION B (40 Marks) – Choose Any Two Questions

Question Two (20 marks)

a. Describe **TWO** features of each of the following types of Local Area Network (LAN):

i. Peer-to-peer network **(4 marks)**

ii. client-server network **(4 marks)**

b. Using examples, differentiate between passive and active attacks **(6 marks)**

c. Describe the following protocols **(6 marks)**

i. Dynamic Host Configuration Protocol

ii. Simple Network Management Protocol

iii. Lightweight Directory Access Protocol

iv. Internet Control Message Protocol

v. Boot Protocol

vi. Point to Point Tunneling Protocol

Question Three (20 marks)

a. Mail servers can be configured with POP or IMAP protocol. What is the difference between the two? **(2 marks)**

b. Discuss the following access methods techniques:

i. CSMA/CD **(4 marks)**

ii. CSMA/CA **(5 marks)**

c. Using illustrations where appropriate, Discuss **THREE** main types of switching techniques as used in computer networks. **(9 marks)**

Question Four (20 marks)

a. Describe any **TWO** computer networks attacks and any **TWO** web attacks **(8 marks)**

b. Using an illustration, describe the FDM and TDM multiplexing **(8 marks)**

c. IEEE designated thinnet and thicknet coax as 10Base2 and 10Base5. What does the notation 10Base2 and 10Base5 indicates? **(4 marks)**

Question Five (20 marks)

a. Using illustrations, Explain the functions of the following network devices

i. *Hub* **(2 marks)**

ii. *Repeater* **(2 marks)**

iii. *Bridge* **(3 marks)**

iv. *Router* **(2 marks)**

v. *Brouter* **(3 marks)**

vi. *Gateway* **(2 marks)**

b. *In ring networks, token passing is used to control the channel access. Using an illustration, describe how token passing works in the network* **(6 marks)**

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KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS

2023/2024 ACADEMIC YEAR

SECOND YEAR FIRST SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

COURSE CODE: COM 211

COURSE TITLE: DATA STRUCTIURES AND ALGORITHMS

DATE: 21ST DECEMBER, 2023 TIME: 2:00-4:00 PM

INSTRUCTION TO CANDIDATES

- SEE INSIDE

Instructions: *Answer ALL Questions in SECTION A and ANY TWO in SECTION B*

QUESTION ONE (1) is COMPULSORY (Carries 30 Marks)

QUESTION 1: (COMPULSORY) 30 MARKS

a) Define the following terms as used in data structures. **(2 Marks)**

i) Space complexity

ii) Time complexity

b) An array *catmarks[]* stores n elements all of integer data type.

i) Write an algorithm that can be used to determine if an element x exists in the array using a linear search technique. **(5 Marks)**

ii) Covert the above algorithm into a program. **(5 Marks)**

c) In order to use a stack efficiently, it is important to always check its status before processing.

i) Identify and explain any three functions that can be used to check the status of a stack. **(3 Marks)**

ii) Design an algorithm for each of the function in b(i) above.

(6 Marks)

d) In order to search a binary tree for a goal node, it is mandatory to traverse it,

i) Identify any three traversal techniques that exist for this purpose.

(3 Marks)

ii) Design the algorithms that can be used for each of the technique identified in (i) above. **(6 Marks)**

SECTION B - ANSWER ANY TWO (2) QUESTIONS (Carries 40 Marks)

QUESTION 2: (20 MARKS)

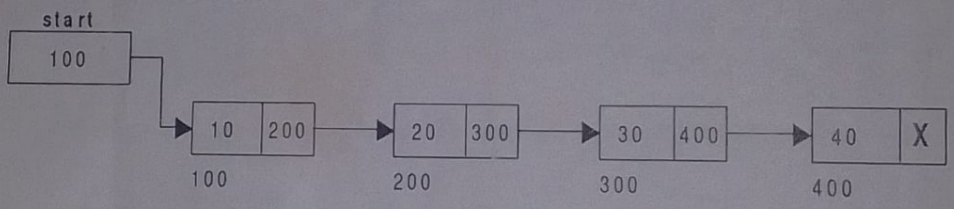
a) The total space occupied by the program during its execution is the sum of the fixed space and the variable space. What's the difference between fixed and variable space? **(2 Marks)**

b) One of the most primitive operations that can be done in a singly linked list is the insertion of a node.

i) Identify and write the syntax for the three functions used to insert a new node at three different places in a singly linked list. **(3 Marks)**

ii) Explain the steps followed to insert a new node at the beginning and the end in a singly linked list. **(6 Marks)**

c) The diagram below shows a single linked list of four nodes. **(9 Marks)**



Write a program for generic function createlist() that can be used to create the four nodes.

QUESTION 3: (20 MARKS)

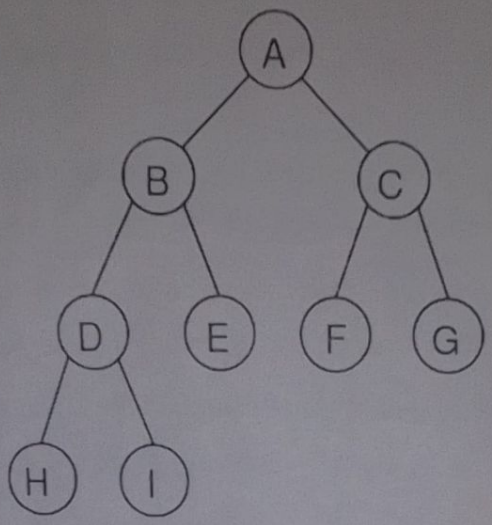
a) Design an algorithm that can be used to insert an item in a linear queue. **(5 Marks)**

b) Write code snippets for implementing the following operations in Queues. **(6 Marks)**

i) enqueue()

ii) dequeue()

c) The diagram below shows a binary tree. Draw a diagram for its linked list representation **(4 Marks)**



d) A linked list is a sequence of data structures, which are connected together via links. Discuss the differences between doubly linked lists and circular linked lists. **(4 Marks)**

e) Using a general syntax, explain the use of the malloc () function in linked lists. **(1 Mark)**

QUESTION 4: (20 MARKS

a) Explain the following terms as used in binary trees. **(3 Marks)**

i) Directed path

ii) Degree of a tree

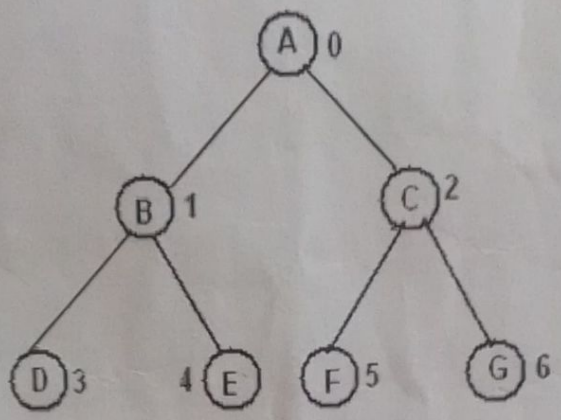
iii) Leaf node

b) Suppose h = height of a binary tree, write expressions to determine the following.

i) Maximum number of leaves. **(1 Mark)**

ii) Maximum number of nodes (1 Mark)

c) Study the binary tree below and use it to answer the questions that follow.



i) Develop its full array representation for the tree. (3 Marks)

ii) Explain how one can identify the following nodes in this tree. (8 Marks)

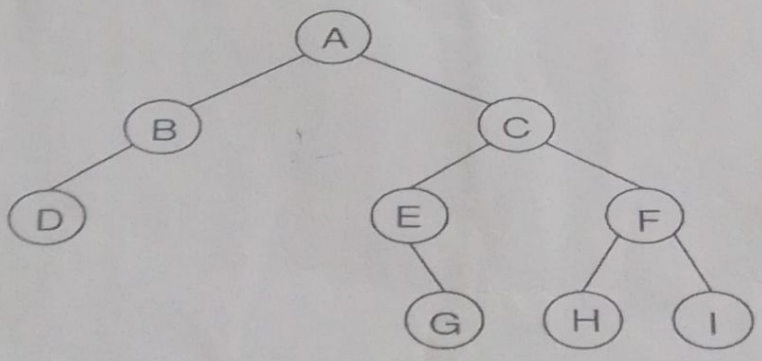
a) Father (n)

b) Lchild (n)

c) Rchild (n)

d) Siblings

d) Study the binary tree below and answer the question that follows.



Indicate what each of the following methods of traversing the tree will yield. **(4 Marks)**

i) Postorder

ii) Preorder

iii) Inorder

iv) Levelorder

QUESTION 5: (20 MARKS)

a) Given 'n' elements which have been ordered by keys so that $x1 < x2 < \ldots < xn$. Write an algorithm that uses the binary search technique to determine if element X exists in the list. **(6 Marks)**

b) Write a complete non recursive program for the binary search algorithm in (a) above. **(8 Marks)**

c) Suppose x is an array of size n stored in memory. Demonstrate using at least three passes how the selection sort algorithm can be used to sort the array in ascending order. **(6 Marks)**

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UNIVERSITY EXAMINATIONS
2022/2023 ACADEMIC YEAR

SECOND YEAR **FIRST** SEMESTER EXAMINATION

FOR THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

COURSE CODE: BIT 216

COURSE TITLE: INTRODUCTION TO COMPUTER NETWORKING

DATE: 13TH DECEMBER 2022 TIME: 9.00 – 12.00PM

INSTRUCTION TO CANDIDATES

SEE INSIDE

INSTRUCTIONS:

Answer **ALL** questions in section A and **ANY THREE** from section B

SECTION A (31 MARKS) - Compulsory

Question One (16 marks)

a. The IT manager in ABC Ltd wants to connect their five PCs to form a WLAN. Explain three disadvantages of this compared to using cabling. (3 marks)

b. Assume that a network used fiber optic cable as the company's backbone cable. Describe three advantages of fiber optic cable over copper cables. (3 marks)

c. Explain any two types of transmission impairments in a network (4 marks)

d. Discuss the three switching techniques (6 marks)

Question Two (15 marks)

a. A _____ is a device that forwards packets between networks by processing the routing information included in the packet. (1 mark)

b. When does network congestion occur? (1 mark)

c. Outline any four reasons as to why a company might prefer a web hosting company to host their website instead of putting it onto a peer-to-peer network (4 marks)

d. Differentiate between:

i. UTP and STP cable (5 marks)

ii. Physical and logical topology (2 marks)

e. Explain why OSI model is called "Open System Interconnection" (2 marks)

SECTION B (39 Marks) – Choose Any Three Questions

Question Three (13 marks)

a. The OSI Reference Model defines seven protocol layers, each of which is responsible for a specific range of functions. By considering this model, explain the main functions of the following layers: (9 marks)

i. The Physical layer

ii. The Data Link layer

iii. The Application layer

b. Give one example of a device on a network that is required to operate in all seven layers of the OSI Reference Model. (1 mark)

c. Explain how data is transmitted along a fiber optic cable. (3 marks)

Question Four (13 marks)

a. Define ADSL (1 mark)

b. Distinguish between TCP and UDP datagram (4 marks)

c. Explain any four advantages of computer networks (8 marks)

Question Five (13 marks)

a. The Institute of Electrical and Electronics Engineers (IEEE) designates Ethernet or thick ethernet cable types as *10Base2* and *10Base5*, respectively, what does the notations *10Base2 or 10Base5* indicate? (4 marks)

b. Using illustrations, describe four network topologies that exist (8 marks)

c. Define the term computer network (1 mark)

Question Six (13 marks)

a. Using a diagram, describe the ATM network (5 marks)

b. Explain the difference between a Virtual Path and a Virtual Channel (4 marks)

c. Write short notes on the following:

i. Terrestrial Microwave (2 marks)

ii. Infrared Transmission (2 marks)

Question Seven (13 marks)

a. Explain how a network connection is established between a user and a server over the Internet using TCP protocol. (5 marks)

b. Discuss four advantages of internet (8 marks)

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UNIVERSITY EXAMINATIONS
2022/2023 ACADEMIC YEAR

SECOND YEAR **FIRST** SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE, BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY, BACHELOR OF SCIENCE IN ACTURIAL SCIENCE AND BACHELOR OF SCIENCE IN STATISTICS

COURSE CODE: COM 211

COURSE TITLE: DATA STRUCTURES AND ALGORITHMS

DATE: 19/12/2022 TIME: 2:00PM-5:00PM

INSTRUCTION TO CANDIDATES

- SEE INSIDE

Instructions: Answer ALL Questions in Section A and Any Three in Section B

SECTION A is COMPULSORY (Carries 31 Marks)

SECTION A (COMPULSORY) 31 MARKS

QUESTION ONE: (COMPULSORY) 16 MARKS

a) Describe the TWO components of a data structure [2 marks]

b) Data structures can be homogenous or non-homogenous. Use examples to support this statement. [4 marks]

c) Explain the time/space trade off as applied in selecting a data structure. [2marks]

d) Explain the two techniques that can be used to represent linear data structures in the computer memory. [2marks]

e) The following values are to be stored in a hash table 9, 14, 29, 67, 123, 184, 251. Describe how the values are hashed by using division method of hashing with a table size of 7. [3 marks]

f) Use chaining as the method of collision resolution. [3 marks]

QUESTION TWO: (COMPULSORY) 15 MARKS

a) Explain the term underflow with respect to linked lists. [2 marks]

b) Explain the following operations performed on data structures. [3 marks]

i. Traversing

ii. Inserting

iii. Deleting

c) Differentiate between queues and stacks. [4 marks]

d) By use of an example in each case, describe the following categories of data structures. [6 marks]

i. Primitive data structures

ii. Basic data structures

iii. Nonlinear data structures

SECTION B – CHOOSE ANY THREE QUESTION (31 MARKS)

QUESTION THREE (13 MARKS)

a) State ONE disadvantage of using an array to implement Queues. [1 mark]
b) Differentiate between merge and sort operations [2 marks]
c) Use quick sort method to sort the following array. [4 marks]

24 56 47 35 10 90 82 31

d) The efficiency of an algorithm can be explained in terms of time and space. Support this statement. [6 marks]

QUESTION FOUR (13 MARKS)

a) Describe any of the properties of binary search algorithms. [1 mark]
b) Describe the following notations as used in time complexity of an algorithm. [3 marks]
   i. Big O
   ii. Theta
   iii. Omega
c) Study the sequence of numbers below and attempt the questions that follow.

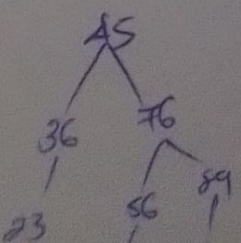
45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 43

   i. Make a Binary Search Tree [3 marks]
   ii. Traverse the tree in preorder [3 marks]
   iii. Traverse the tree in postorder [3 marks]

QUESTION FIVE (13 MARKS)

a) Describe the following terms as used in a tree data structure [3 marks]
   i. Root
   i. Leaf
   ii. Parent
b) Describe the following terms as used in data structures and algorithms. [4 marks]
   i. Recursion
   ii. Heap
c) Convert the following infix expressions into prefix and postfix expressions [6 marks]
   i. A-(B-(C-D))
   ii. A-B-C-D
   iii. (A+B)*(C/(D-E))

QUESTION SIX (13 MARKS)



a) Explain the following types of graphs. [3 marks]

i. Mixed
ii. Undirected
iii. Directed

b) Differentiate between double ended queue and circular linked list. Use illustrations. [4 Marks]

c) A queue is implemented using ADT list to presents its items. Discuss the efficiency of the queue insertion and deletion operations when the ADT list implementation is:

i. Array based [3 marks]
ii. Pointer based [3 marks]

QUESTION SEVEN (13 MARKS)

a) Explain three strategies that divide and conquer algorithms use to solve problems [3 marks]

b) Describe four real world application of algorithms [4 marks]

c) Write Pseudo code for each of the following operations of linked lists. [6 marks]

i. Inserting a node
ii. Traversing a linked list

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UNIVERSITY EXAMINATIONS

2023/2024 ACADEMIC YEAR

SECOND YEAR **FIRST** SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

COURSE CODE: COM 211

COURSE TITLE: DATA STRUCTIURES AND ALGORITHMS

DATE: 21ST DECEMBER, 2023 TIME: 2:00-4:00 PM

INSTRUCTION TO CANDIDATES

- SEE INSIDE

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS

2020/2021 ACADEMIC YEAR

SECOND YEAR FIRST SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE/
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

COURSE TITLE: COM 213

COURSE CODE: DATABASE SYSTEMS

DATE: 14TH DECEMBER, 2020 TIME: 2:00-5:00 PM

INSTRUCTION TO CANDIDATES

- SEE INSIDE

Instructions: Answer ALL Questions in Section A and Any Three in Section B

SECTION A: 31 MARKS (COMPULSARY)

QUESTION ONE (16 MARKS)

a) Differentiate between the following terms [6 marks]

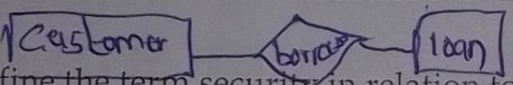
i) Simple and composite attributes

ii) Meta-data and user data

iii) Primary key and Super key

b) Explain any **THREE** benefits a Bank is likely to get by adopting the use of a database system in its day to day operations [6 marks]

c) Use an ER diagram to illustrate a relationship set borrower to denote the association between customers and the bank loans that the customers have. [3 marks]



d) Define the term security in relation to database systems [1 mark]

QUESTION TWO (15 MARKS)

a) Use diagrams to illustrate the following database systems [4marks]

i) Distributed database system

ii) Centralized database systems

b) Give **FOUR** reasons as to why one would choose the distributed database systems over the centralize database systems. [4 marks]

c) A proposed database for a DVD order processing system will have four relations as listed below.

- Customers
- Customer orders
- Order details
- DVD

Required

i) Derive the fields for the relations above [4 marks]

ii) Suggest the key fields for each table [3 marks]

SECTION B: ANSWER ANY THREE QUESTIONS (39 MARKS)

QUESTION THREE (13 MARKS)

a) i) Define the term database management system? Give 2 examples. [3 marks]

ii) Explain any **FIVE** functions of the database management system. [5 marks]

b) Use examples to discuss the **FIVE** major components of a database systems environment. [5 marks]

QUESTION FOUR (13 MARKS)

a) The human resource manager has requested you to come up with a job description of database administrator (DBA). Discuss any **FIVE** functions of a DBA that you will propose for adoption. [5 marks]

b) Describe the three-schema database architecture, outlining the different mappings between the schema levels [8 marks]

QUESTION FIVE (13 MARKS)

a) List **FIVE** reasons as to why there's need to protect the database [5 marks]

b) Identify and state the purpose of any FOUR Microsoft Access database objects [4 marks]

c) Consider an entity relationship subsisting between a student and a course. Generate a sample tabular representation of the above relationship, assuming key attributes are course-code and Reg-no respectively. [4 marks]

QUESTION SIX (13 MARKS)

a) Using diagrams describe the following data models. [9 marks]

i) Network

ii) Relational

iii) Hierarchical

b) Differentiate between the following concepts as used in database systems [4 marks]

i) Entity type and entity set

ii) Meta-data and User-data

QUESTION SEVEN (13 MARKS)

a) Differentiate between the following.

i) Database state and database schema [4 marks]

ii) Physical data independence and logical data independence

b) Describe the **SIX** clauses in the syntax of an SQL query. [6 marks]

c) Using diagrams explain any **THREE** types of table relationships. [3 marks]

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KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS
2021/2022 ACADEMIC YEAR

SECOND YEAR FIRST SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY

COURSE CODE: COM 210

COURSE TITLE: STRUCTURED PROGRAMMING

DATE: 22ND FEBRUARY 2022 TIME: 9.00-12.00PM

INSTRUCTION TO CANDIDATES

- SEE INSIDE

INSTRUCTIONS: *Answer ALL Questions in SECTION A and ANY THREE in SECTION B*

SECTION A is COMPULSORY - CARRIES 31 MARKS

QUESTION ONE (16 MARKS)

a) Use an illustration to demonstrate the difference between source code and object code in C programming (3 Marks)

b) Consider the code snippet below and determine the value of c after the code executes (4 Marks)

```
int [] a = {1, 2, 3, 4, 5};
int [] b = {11, 12, 13};
int [] c = new int[4];
for (int j = 0; j < 2; j++)
{
c[j] = a[j] + b[j];
}
```

c)

i) What is a function prototype and when are they used? (3 Marks)

ii) Explain the meaning of the following function prototype:

void test(long a, short b, unsigned c); (3 Marks)

d) Write a C program to demonstrate the difference in sizes of a long and short as reserved in memory (3 Marks)

QUESTION TWO (15 MARKS)

a)

i) Write a code snippet that declares a two-dimensional array to store characters. (3 Marks)

ii) Show how the above array would be initialized with some data interactively during runtime. (4 Marks)

b) Use the data in the table below to develop a C program that uses the switch structure (5 Marks)

Day of the week	Morning Activity	Afternoon Activity
Monday	Jogging	Bike riding
Wednesday	Gym	Jogging
Friday	Indoor exercises	Nature walk
Sunday	Church	Movie time

c) Explain the THREE types of preprocessor directives that can be embedded in a C program (3 Marks)

SECTION B

(SECTION B CARRIES 39 MARKS - CHOOSE ANY THREE QUESTIONS)

QUESTION THREE (13 MARKS)

a) Write a C program that mimics a Loan Calculator that receives the principle amount and desired repayment period from a user then determines the following; (10 Marks)

i) The amount payable as interest given that the bank charges 13.4% per annum;

ii) The amount payable per month

iii) The total amount paid at the end of the loan term

iv) The processing fee given that the bank charges a processing fee of 2.7% of the principle amount.

b) Use a code snippet to demonstrate two ways of initializing and calling a constant in C. (3 Marks)

QUESTION FOUR (13 MARKS)

a) Use a code snippet to demonstrate the use of getter and setters in a C program (6 Marks)

b) Define a pointer and state their roles in C programing (3 Marks)

c) Use an illustration to show how structs differ from unions in C (4 Marks)

QUESTION FIVE (13 MARKS)

a) List and explain the FOUR main C storage classes (6 Marks)

b) Write a suitable C program using a structure to hold the following details and which defines a new type as shown below: (7 Marks)

Element	Type
Car_Name	char (20)
Model	char(50)
Transmission	char(30)
YoM	int
Sold	bool

QUESTION SIX (13 MARKS)

a)

i) What is an enumerated type? (1 Mark)

ii) Define an enumerated type to hold the following possible values on user interface capturing bio data: married, single, divorced, widowed. Show how a variable can be used with this enumerated type. (6 Marks)

b)

i. Write a program to calculate and display the average of the odd numbers between 0 and 333 inclusive, using a do while loop. (3 Marks)

ii. Convert the program in question b(i) above such that it uses a for loop to display the even numbers in the same range (3 Marks)

QUESTION SEVEN (13 MARKS)

a) Use an illustration to show how escape sequences differ from format specifiers (2 Marks)

b) Describe the following file opening modes: "r","a","w". (3 Marks)

c) Write a program that will open a file called TEST.TXT and write the text "Welcome to Structured Programming" to it. (3 Marks)

d) Write a C program to demonstrate how recursive functions work (5 Marks)

9/15

Name Lilian Wangui Karonga

Reg no. P102/20616/21

Unit : Structured Programming (Com 210)

Date 6/10/2022

CAT 1

1 How int main () differ from void main ()

An int main () has a return 0 while void main () does not have a return value

2 Ways of initializing a constant discount

Int CONST discount = value;

#define CONST discount = value;

3 C program to test and display results of

a == b && c > b equals to 0

a != b || c < b equals to 1

!a == b equals to 0

```
#include <stdio.h> // preprocessor directives
int main () // Main function
{ // declaring and initializing variables
int a = 15;
int b = 15;
int c = 10;
int d = a == b && c > b;
int e = a != b || c < b;
int f = !a == b; equals to
// giving the outputs
Printf("The output for variable d is: %d", d);
// the output gives a 0
Printf("The output for e is: %d", e);
// the output gives 1
```

```
Printf("The output for f is: %d", f);
    // the output equals to 0
    return 0;
}
```

4 A Programming statement that displays good morning if the time (in hours) is before midday and displays good day if the time (in hours) is past midday

```
Time ? good_morning : good-day ;
```

5 C Program that converts hours to minutes and seconds

```
#include<stdio.h>
int main()
{       // declaration and initialization of variables
    int hours;
    int minutes;
    int seconds;
Printf("Hours: %d\t", hours *360 );
Printf("Minutes: %d\t", Minutes * 60);
Printf("Seconds: %d\t", seconds);
    return 0;
}
```

Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS
2021/2022 ACADEMIC YEAR

SECOND YEAR FIRST SEMESTER EXAMINATION

FOR THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY/COMPUTER SCIENCE & BACHELOR OF SCIENCE

COURSE CODE: BIT 216

COURSE TITLE: INTRODUCTION TO COMPUTER NETWORKS

DATE: 21st FEBRUARY 2022 TIME: 9.00 – 12.00 PM

INSTRUCTIONS TO CANDIDATES

SEE INSIDE

INSTRUCTIONS:

Answer **ALL** questions in section **A** and **ANY THREE** from section **B**

SECTION A (31 MARKS) –Compulsory

QUESTION ONE (16 MARKS)

a. What is a network protocol? Set of rules governing communication in a network. (2 marks)

b. Using a well labeled diagram, describe the implementation of VOIP and how it works (6 marks)

c. Describe the terms confidentiality, integrity and authentication (3 marks)

d. Explain the function of DNS system (1 mark)

e. Highlight FOUR advantages of ring topology over other topologies (4 marks)

QUESTION TWO (15 MARKS)

a. Discuss the role of the transport layer in the TCP/IP protocol stack (3 marks)

b. Assume that you have been requested by an organization with ten (10) computers to instal their LAN where five (5) computers will be connected through wired connection while fiv (5) will be connected via wireless connectivity. Draw a well labelled design of the networ that you will implement for the organization and explain how the wireless connection wi be established. Use bus topology (5 marks)

c. Twisted pair cables are the mostly widely used cables in LANs.

i. Describe how twisted pair cables are designed to reduce EMIs, crosstalk and attenuation. (2 marks)

ii. List any TWO benefits and any THREE drawbacks of using twisted pair cable (5 marks)

SECTION B (39 Marks) - Choose Any Three Questions

QUESTION THREE (13 MARKS)

a. Describe any **FIVE** protocols applicable to TCP/IP protocol suite (10 marks)

b. The Institute of Electrical and Electronics Engineers (IEEE) designates Ethernet or thick ethernet cable types as *10Base2* and *10Base5*, respectively, what does the notations *10Base2 or 10Base5* indicate? (3 marks)

QUESTION FOUR (13 MARKS)

a. As an ICT technician, you were requested to connect first year student laptop to the network. Explain the steps you will follow toconfigure their machines to connect to the internet (5 marks)

b. State **SIX** differences between UTP and STP cable (6 marks)

c. Describe the terms Physical and logical topology as used in computer networks (2 marks)

QUESTION FIVE (13 MARKS)

a. The OSI Reference Model defines seven protocol layers, each of which is responsible for a specific range of functions. By considering this model, explain the main functions of the following layers: (9 marks)

   i. Physical layer

   ii. Data link layer

   iii. Application layer

b. Give one example of a device on a network that is required to operate in all seven layers of the OSI Reference Model. (1 mark)

c. Explain how data is transmitted along a fiber optic cable. (3 marks)

QUESTION SIX (13 MARKS)

a. Using a diagram, describe the ATM network (5 marks)

b. What is the difference between a Virtual Path and a Virtual Channel? (4 marks)

c. Write short notes on the following:

i. Terrestrial Microwave (2 marks)

ii. Infrared Transmission (2 marks)

QUESTION SEVEN (13 MARKS)

a. Explain how a network connection is established between a user and a server over Internet using the TCP protocol. (5 marks)

b. Discuss FOUR advantages of the internet (8 marks)

CAT II COM 210 21ST NOVEMBER 2022

1. Consider a variable called temp which stores the current temperature as received from a user. Write a C program that uses a pointer to hold and display the address of the variable temp.
2. Write a C program that receives a number from a user, displays it and then shows the previous and next number
3. Use a Struct to create a C program which can be used to hold the information of a customer who has an account at KCB Bank
4. Write a C program that asks a user to enter their name and stores the name in binary file called biodata.txt
5. Show how a function would be used in C program to issue a discount of 3% on the total sale above 300 and display the discount issued.

COM 210: Structured Programming CAT I 6TH OCTOBER 2022

1. How does int main() differ from void main() (2 Marks)
2. Show two ways a constant called discount can be initialized (2 Marks)
3. Write a C Program to test and display results as shown below (5 Marks)

   a == b && c > b equals to 0

   a != b || c < b equals to 1

   ! a == b equals to 0
4. Using one programming statement, write a code snippet to that would display good morning if the time (in hours) is before midday and displays good day if the time (in hours) is past midday (3 Marks)
5. Write a C program to convert a given integer (in seconds) to hours, minutes and seconds. The output should be as shown below (3 Marks)

Hours: Minutes: Seconds:

Seconds *

printf("Hours: %d",

~~good Morning : good day~~;